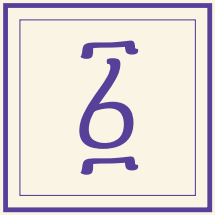
GEPA

Reflective prompt evolution (Agrawal et al., ICLR 2026)

Carlos Cotrini
AI in Industry, ETH Zurich
Weekend 1, October 2026



Section 1

The problem

GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

The answer on file

140

Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

A: The instruction goes here Then the working, then a number

At the start of the answer

The answer on file

140

Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

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The answer on file

140

Right if they match

Yang et al. 2024, Fig. 3 and section 5

GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

A: The instruction goes here Then the working, then a number

At the start of the answer

The answer on file

140

Right if they match



7,473 training problems

To optimise on (a 3.5% sample)



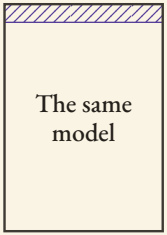
1,319 test problems

To check the result on

Yang et al. 2024, Fig. 3 and section 5

Same model, different instruction

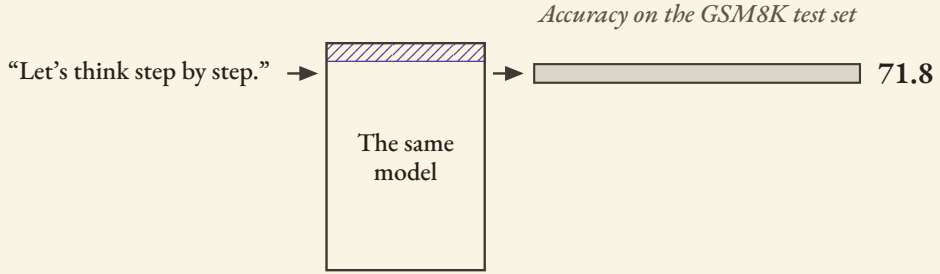
Accuracy on the GSM8K test set



PaLM 2-L

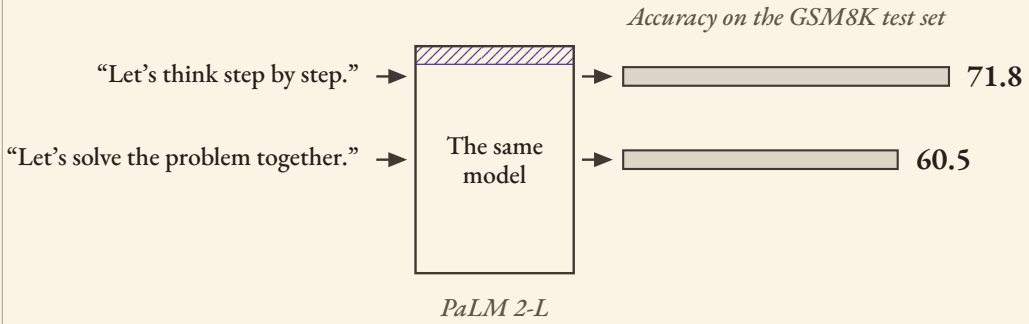
Yang et al. 2024, section 5.2.3, p. 13

Same model, different instruction



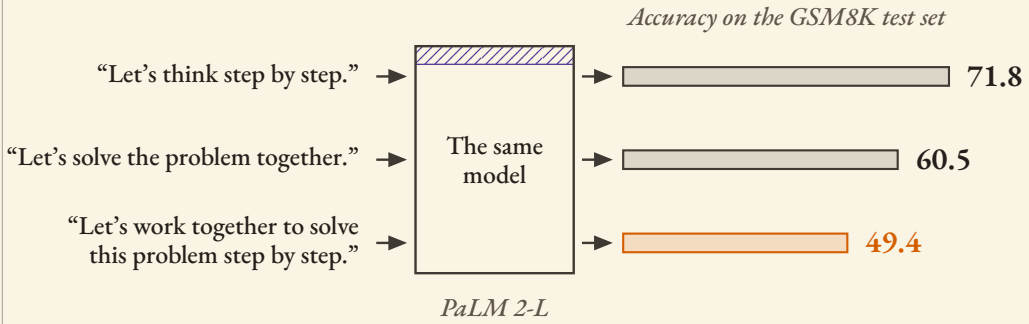
Yang et al. 2024, section 5.2.3, p. 13

Same model, different instruction



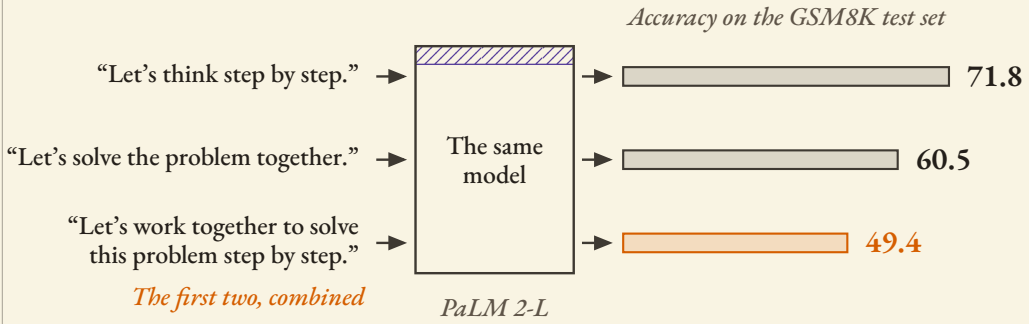
Yang et al. 2024, section 5.2.3, p. 13

Same model, different instruction



Yang et al. 2024, section 5.2.3, p. 13

Same model, different instruction



Yang et al. 2024, section 5.2.3, p. 13

One instruction, before and after GEPA

Agrawal et al. 2026, Fig. 2, Table 2

The seed instruction

`create_query_hop2`

Given the fields `question`, `summary_1`, produce the fields `query`.

One instruction, before and after GEPA

The seed instruction

Agrawal et al. 2026, Fig. 2, Table 2

`create_query_hop2`

Given the fields `question`, `summary_1`, produce the fields `query`.

`create_query_hop2` *GEPA's instruction, GPT-4.1 Mini*

[...] **Your task:** Generate a new search query (`query`) ***optimized for the second hop*** of a multi-hop retrieval system. [...]

Key Observations and Lessons:

- First-hop documents often cover one entity or aspect.
- Remaining relevant documents often involve connected or higher-level concepts mentioned in `summary_1` but not explicitly asked in the original question. [...]
- If `summary_1` describes a population for a small civil parish, but the question wants the total population of the wider region, your query should target that wider region (e.g., "Madeira archipelago population in 2011"). [...]

One instruction, before and after GEPA

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Lessons written by the reflecting model

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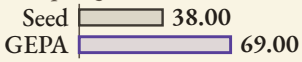
`create_query_hop2` *GEPA's instruction, GPT-4.1 Mini*

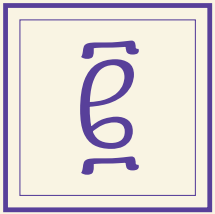
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System's test score, HotpotQA, GPT-4.1 Mini





Section 2

Learning from text

A rollout returns a score and text

`system_prompt` *The seed instruction*

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

A rollout returns a score and text

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You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

Problem x (2022 AIME I, problem 6)
Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a, b, 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

A rollout returns a score and text

`system_prompt` *The seed instruction*

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

Execution trace of GPT-4.1 nano, last line

```
### <final answer> 234
```

Problem x (2022 AIME I, problem 6)
Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a, b, 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.
Gold answer m: `### 228`
GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

A rollout returns a score and text

`system_prompt` *The seed instruction*

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

Problem x (2022 AIME I, problem 6)
Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a, b, 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.
Gold answer m: `### 228`

GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7

→

Execution trace of GPT-4.1 nano, last line

`### <final answer> 234`

The metric: is `### 228` in the response? No: score 0

A rollout returns a score and text

`system_prompt` *The seed instruction*

You are a helpful assistant. You are given a question and you need to answer it. The answer should be given at the end of your response in exactly the format '### <final answer>'

Problem x (2022 AIME I, problem 6)

Find the number of ordered pairs of integers (a, b) such that the sequence 3, 4, 5, a, b, 30, 40, 50 is strictly increasing and no set of four (not necessarily consecutive) terms forms an arithmetic progression.

Gold answer m: `### 228`

Execution trace of GPT-4.1 nano, last line

```
### <final answer> 234
```

The metric: is `### 228` in the response? No: score 0

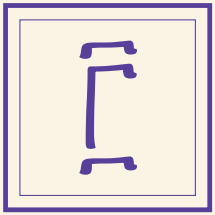
The feedback function returns the score and this text:

Evaluation trace: what the metric wrote

`Feedback`

The generated response is incorrect. The correct answer is '### 228'. Ensure that the correct answer is included in the response exactly as it is. Here is some additional context that might be helpful: solution: [...]

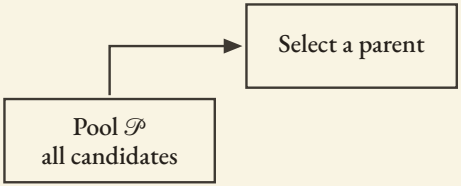
GEPA's recorded AIME test; Agrawal et al. 2026, pp. 6, 7



Section 3

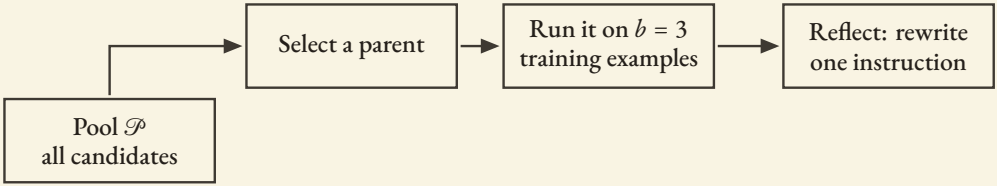
The GEPA loop

GEPA's loop



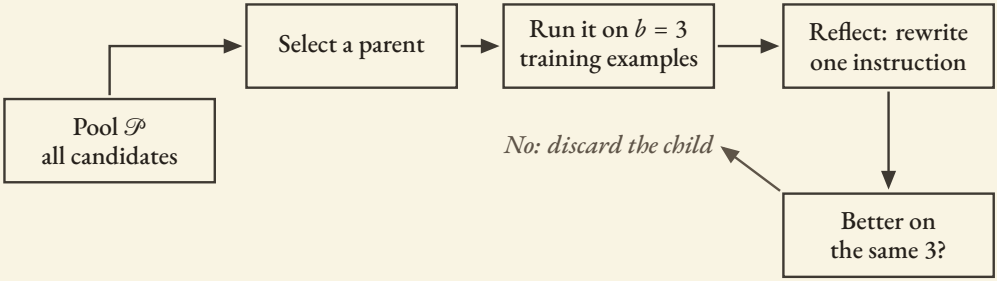
Agrawal et al. 2026, Fig. 3 and Alg. 1

GEPA's loop



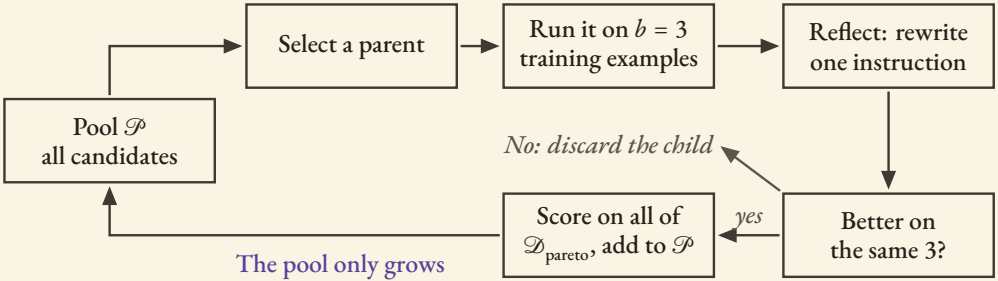
Agrawal et al. 2026, Fig. 3 and Alg. 1

GEPA's loop



Agrawal et al. 2026, Fig. 3 and Alg. 1

GEPA's loop



Agrawal et al. 2026, Fig. 3 and Alg. 1

Validation set: $\mathcal{D}_{\text{pareto}}$

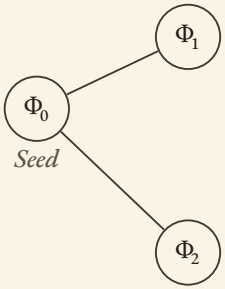
GEPA grows a tree of candidates



Seed

Agrawal et al. 2026, Alg. 1 and Fig. 6

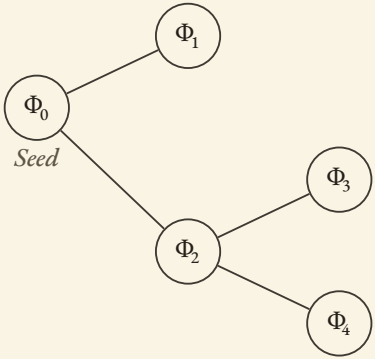
GEPA grows a tree of candidates



Each child is its parent with one instruction rewritten after reflection.

Agrawal et al. 2026, Alg. 1 and Fig. 6

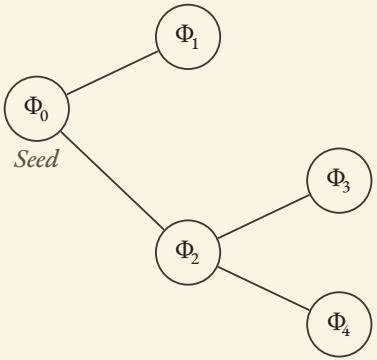
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GEPA grows a tree of candidates

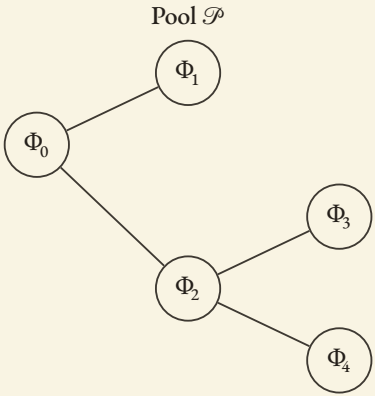


Each child is its parent with one instruction rewritten after reflection.

Pool $\mathcal{P}$:
all the candidates

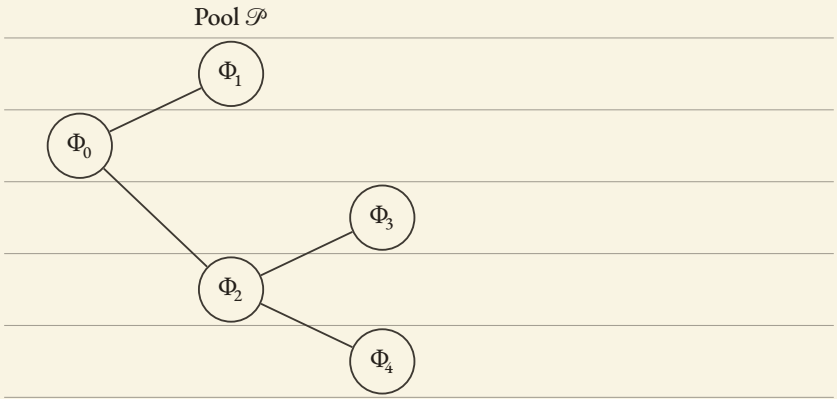
Agrawal et al. 2026, Alg. 1 and Fig. 6

The Pareto front of the pool



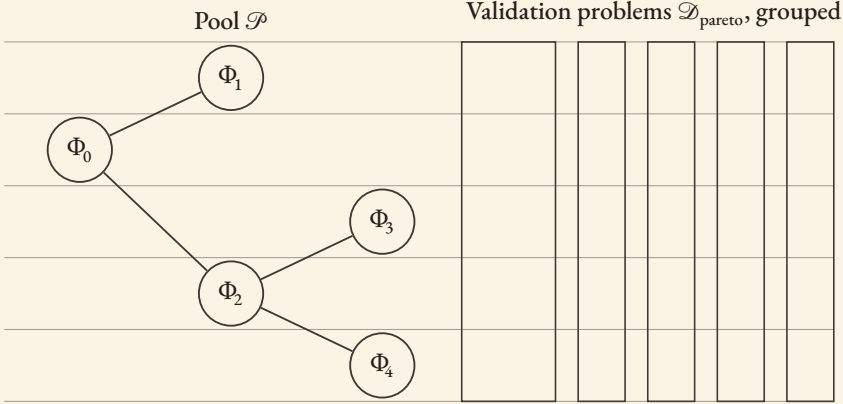
Agrawal et al. 2026, p. 7 and Alg. 2

The Pareto front of the pool



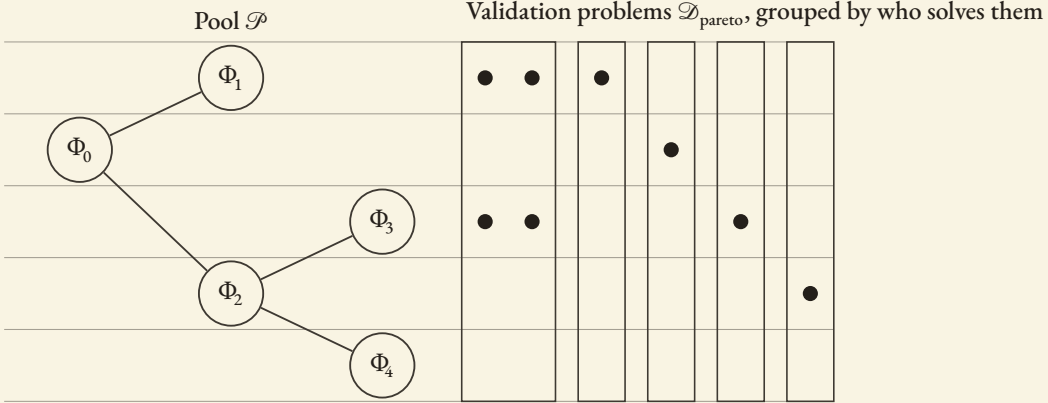
Agrawal et al. 2026, p. 7 and Alg. 2

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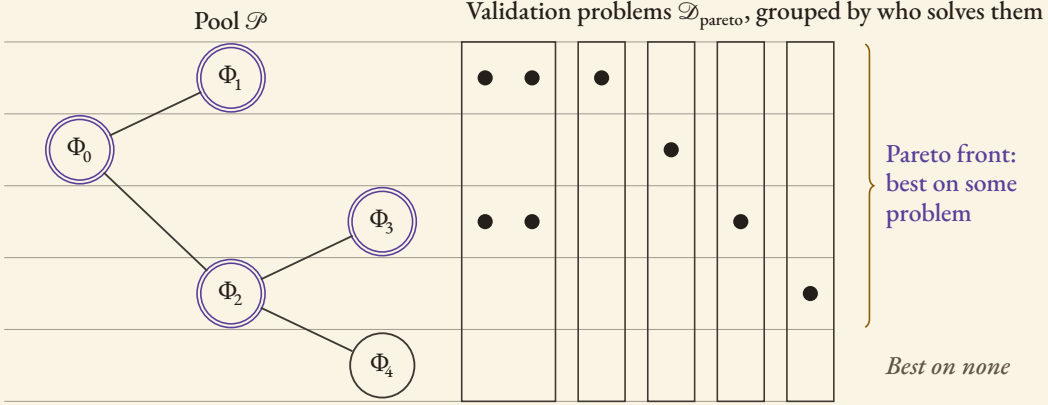
Agrawal et al. 2026, p. 7 and Alg. 2

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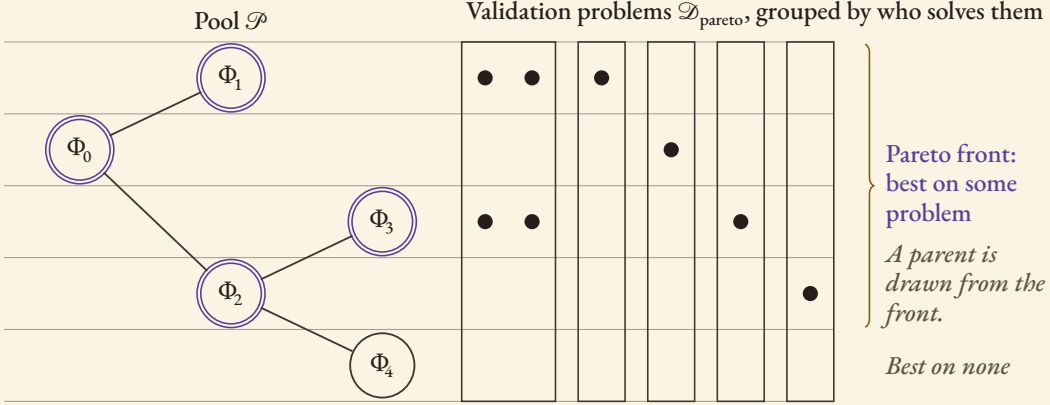
Agrawal et al. 2026, p. 7 and Alg. 2

The Pareto front of the pool



Agrawal et al. 2026, p. 7 and Alg. 2

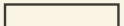
The Pareto front of the pool



Agrawal et al. 2026, p. 7 and Alg. 2

Setup

HotpotQA, questions

$\mathcal{D}_{\text{feedback}}$ *Training split*  150

Minibatches of $b = 3$
Reflection reads inputs, outputs, feedback

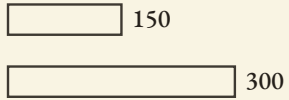
Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Setup

HotpotQA, questions

$\mathcal{D}_{\text{feedback}}$ *Training split* — 150

$\mathcal{D}_{\text{pareto}}$ *Validation split* — 300



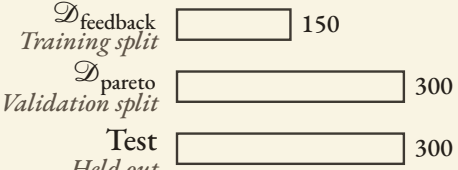
Minibatches of $b = 3$
Reflection reads inputs, outputs, feedback

Each accepted child runs on all
Scores only, to select candidates

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Setup

HotpotQA, questions



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Reflection reads inputs, outputs, feedback

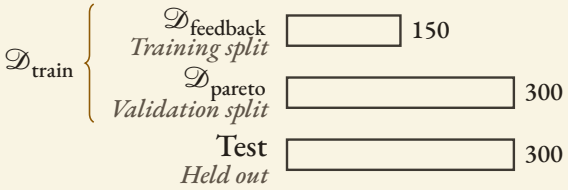
Each accepted child runs on all
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Reported once, at the end

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Setup

HotpotQA, questions



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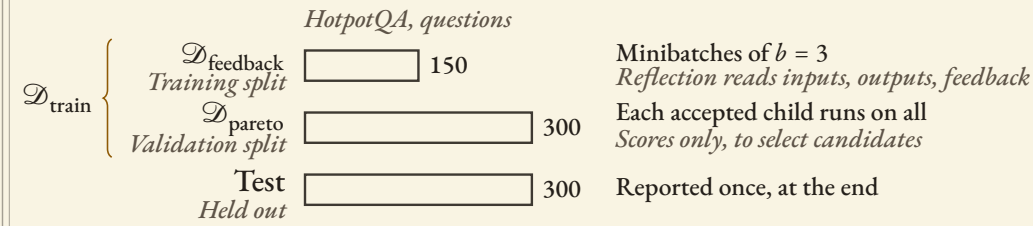
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Reported once, at the end

1 **Split $\mathcal{D}_{\text{train}}$ into $\mathcal{D}_{\text{feedback}}$, $\mathcal{D}_{\text{pareto}}$, s.t. $|\mathcal{D}_{\text{pareto}}| = n_{\text{pareto}}$**
2 Initialize candidates $\mathcal{P} \leftarrow [\Phi]$
3 **for each** (x_i, m_i) **in** $\mathcal{D}_{\text{pareto}}$ **do**
4 $\quad S_\Phi[i] \leftarrow$ the score of Φ on (x_i, m_i)

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Setup



1 Split $\mathcal{D}_{\text{train}}$ into $\mathcal{D}_{\text{feedback}}$, $\mathcal{D}_{\text{pareto}}$, s.t. $|\mathcal{D}_{\text{pareto}}| = n_{\text{pareto}}$

▸ 2 **Initialize candidates $\mathcal{P} \leftarrow [\Phi]$** *// Φ: the seed*

▸ 3 **for each (x_i, m_i) in $\mathcal{D}_{\text{pareto}}$ do** *// all 300 questions*

▸ 4 $S_\Phi[i] \leftarrow$ the score of Φ on (x_i, m_i) *// S_Φ: one row of scores*

Agrawal et al. 2026, Alg. 1, pp. 8, 23, 27

Algorithm 1 in words

Algorithm 1, GEPA

- **while the budget B is not exhausted do**
  - **$\Phi \leftarrow$ select a candidate from the pool $\mathcal{P}$**

  $\mathcal{M} \leftarrow$ sample a few questions from $\mathcal{D}_{\text{feedback}}$
  Use Φ to solve $\mathcal{M}$. Collect feedback F, score σ, traces
  $\Phi' \leftarrow$ update Φ with F, σ and the traces
  Get the score σ' of Φ' on $\mathcal{M}$
  if $\sigma' > \sigma$ **then**
  Add Φ' to $\mathcal{P}$ as a child of Φ

Algorithm 1 in words

Algorithm 1, GEPA

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Algorithm 1 in words

Algorithm 1, GEPA

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Algorithm 1 in words

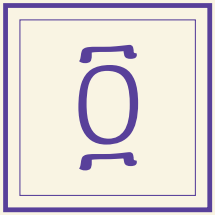
Algorithm 1, GEPA

while the budget B is not exhausted **do**
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  Use Φ to solve $\mathscr{M}$. Collect feedback F, score σ, traces
  $\Phi' \leftarrow$ update Φ with F, σ and the traces
- **Get the score σ' of Φ' on $\mathscr{M}$**
- **if $\sigma' > \sigma$ then**
    Add Φ' to $\mathscr{P}$ as a child of Φ

Algorithm 1 in words

Algorithm 1, GEPA

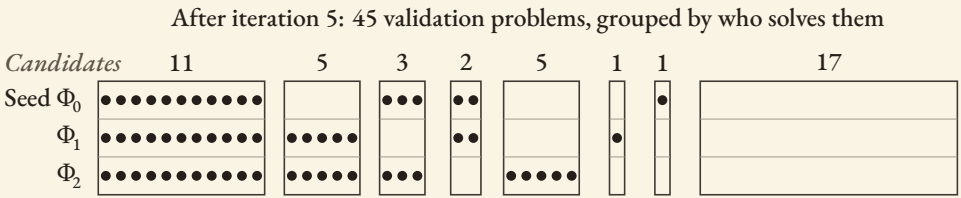
while the budget B is not exhausted **do**
  $\Phi \leftarrow$ select a candidate from the pool $\mathscr{P}$
  $\mathscr{M} \leftarrow$ sample a few questions from $\mathscr{D}_{\text{feedback}}$
  Use Φ to solve $\mathscr{M}$. Collect feedback F, score σ, traces
  $\Phi' \leftarrow$ update Φ with F, σ and the traces
  Get the score σ' of Φ' on $\mathscr{M}$
  if $\sigma' > \sigma$ **then**
    ▸ **Add Φ' to $\mathscr{P}$ as a child of Φ**



Section 4

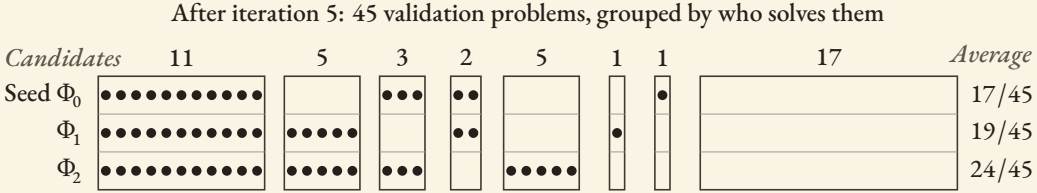
Choosing the parent

The score matrix of a real run



DSPy tutorial "GEPA for AIME": GPT-4.1 Mini solves, GPT-5 reflects

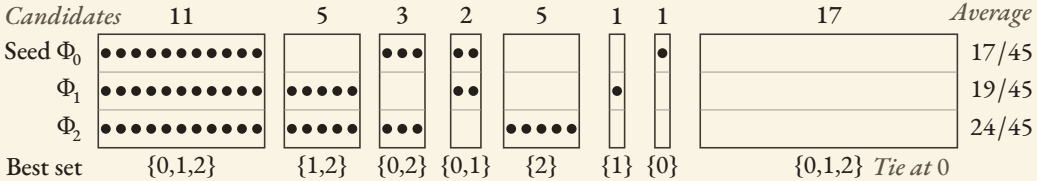
The score matrix of a real run



DSPy tutorial "GEPA for AIME": GPT-4.1 Mini solves, GPT-5 reflects

The score matrix of a real run

After iteration 5: 45 validation problems, grouped by who solves them

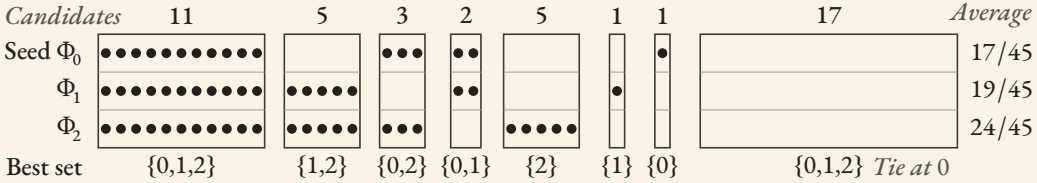


Φ is best on problem *i* when
no candidate scores higher on *i*

DSPy tutorial "GEPA for AIME": GPT-4.1 Mini solves, GPT-5 reflects

The score matrix of a real run

After iteration 5: 45 validation problems, grouped by who solves them



Φ is best on problem *i* when no candidate scores higher on *i*

Dominance: as high on every problem, and higher on one

DSPy tutorial "GEPA for AIME": GPT-4.1 Mini solves, GPT-5 reflects

Algorithm 2 in words

Algorithm 2, Pareto-based candidate selection

- **function SelectCandidate($\mathcal{P}$)**
  - **$\mathcal{C} \leftarrow \emptyset$**
  - **for each problem q in the validation set $\mathcal{D}_{\text{pareto}}$ do**
    - **Add to $\mathcal{C}$ all Φ in the pool $\mathcal{P}$ that best answer q**

  Remove from $\mathcal{C}$ any Φ dominated by another. What is left: $\hat{\mathcal{C}}$

  For Φ in $\hat{\mathcal{C}}$, let $f(\Phi)$ be the number of problems where Φ gives the best answer

  Sample Φ from $\hat{\mathcal{C}}$ with probability proportional to $f(\Phi)$

Algorithm 2 in words

Algorithm 2, Pareto-based candidate selection

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GEPA against GRPO and MIPROv2, per task

Qwen3 8B, test score (%)

	HotpotQA	IFBench	HoVer	PUPA	AIME 2025	LiveBench Math	Aggregate
Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23

Agrawal et al. 2026, Table 1; Opsahl-Ong et al. 2024

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Rollouts, GRPO	24,000	24,000	24,000	24,000	24,000	24,000	

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Instructions and examples, chosen by Bayesian search							
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GEPA	62.33	38.61	52.33	91.85	32.00	51.95	54.85
Rollouts, GRPO	24,000	24,000	24,000	24,000	24,000	24,000	
Rollouts, GEPA	6,871	3,593	7,051	2,426	1,839	1,839	

Agrawal et al. 2026, Table 1; Opsahl-Ong et al. 2024

GEPA against GRPO and MIPROv2, per task

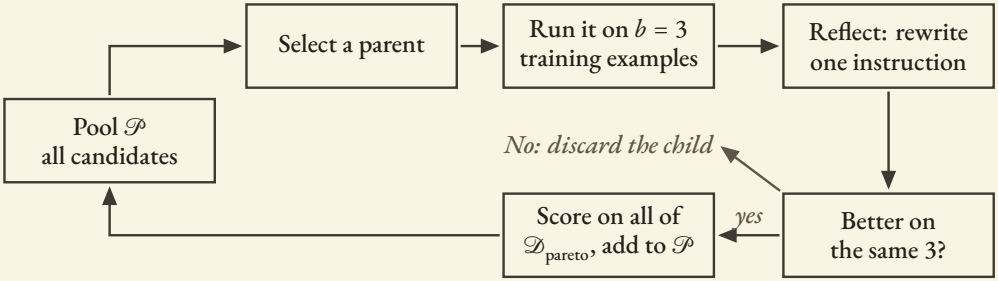
Qwen3 8B, test score (%)

GRPO ahead: 38.00 against 32.00

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GEPA, assembled

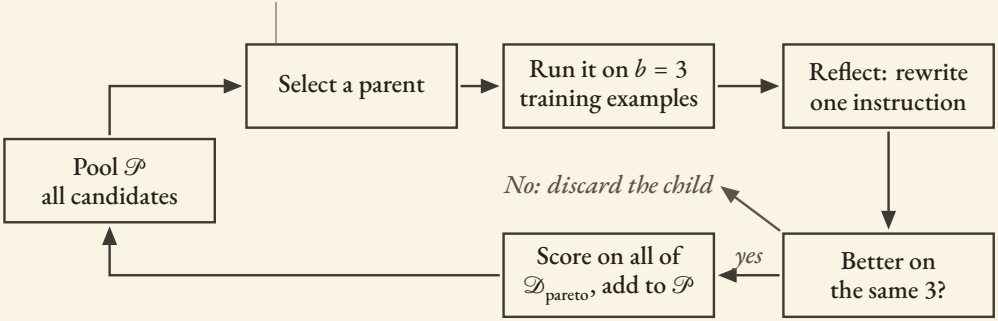


Agrawal et al. 2026, Fig. 3 and Alg. 1 and 2

Validation set: $\mathcal{D}_{\text{pareto}}$

GEPA, assembled

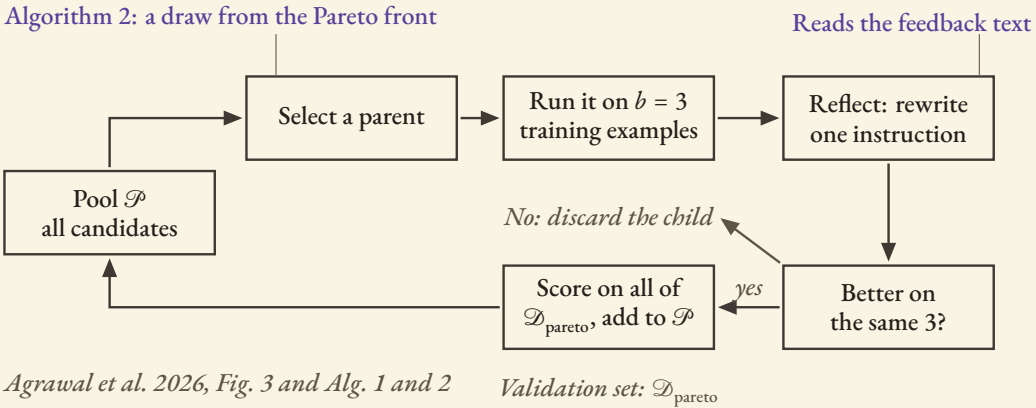
Algorithm 2: a draw from the Pareto front



Agrawal et al. 2026, Fig. 3 and Alg. 1 and 2

Validation set: $\mathcal{D}_{\text{pareto}}$

GEPA, assembled



Agrawal et al. 2026, Fig. 3 and Alg. 1 and 2

Validation set: $\mathcal{D}_{\text{pareto}}$

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Runs quoted in this lecture: the GEPA library's recorded AIME test, `tests/test_aime_prompt_optimization` at commit `d771eb2`; the DSPy tutorials "GEPA for AIME", a run of 12 August 2025, and "GEPA for Privacy-Conscious Delegation", dspy.ai/tutorials/gepa_aime and dspy.ai/tutorials/gepa_papillon